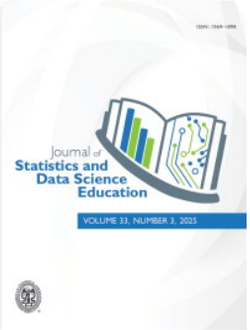

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The Impact of Homework Deadline Times on College Student Performance and Stress: A Quasi-Experiment in Business Statistics

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ABSTRACT

Homework is vital to undergraduate student learning, but college students also rank it as one of their biggest stressors. Online homework deadline times may affect student learning outcomes and mental health. Particularly, the de facto standard of 11:59 p.m. potentially contributes to a lack of sleep and may encourage waiting until the night an assignment is due. Yet little research has experimentally investigated the effect of homework deadline times. This article presents a quasi-experiment in a business statistics course where different sections are randomly assigned a 4 p.m. or 11:59 p.m. homework deadline. Results from surveys (n= 84) and sentiment analysis of course evaluations indicate students prefer 11:59 p.m. deadlines and report less stress from them. However, these perceptions do not translate into significant differences in learning outcomes. The study suggests avenues for further research on homework deadlines involving different methodologies and situations. Supplementary materials for this article are available online.

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College teaching; Homework deadlines; Online homework; Student stress

1. Introduction

An educator from Texas posted a meme on LinkedIn saying, "Every college student knows the importance of 11:59," and added a comment, "A little college humor that is ABSOLUTELY true." Within a month, the post had received 44,583 likes, 1206 comments, and 1262 reposts (Barone 2022). That viral response illustrates how much online homework deadline times impact the modern college student experience.

Altering those deadlines is one tool college instructors can use when trying to balance homework's positive learning benefits with its adverse side effects. Too many out-of-class assignments can overwhelm students as they juggle multiple courses and college social life, as well as other non-school obligations like part-time jobs and family obligations. Course workload is the most frequently mentioned challenge among statistics students (Sutter et al. 2023). This potentially increases student stress, which has ballooned since the COVID-19 pandemic (Lukowski et al. 2024). Dr. Sarah K. Lipson, an author of a national study of more than 30,000 college students' mental health during 2020, suggests a 5 p.m. homework deadline rather than midnight or 9 a.m. because those times encourage students to go to bed later, which is associated with many mental health problems (McAlpine 2021).

This article introduces a quasi-experiment where sections of the same business statistics course (Quantitative Business Analysis) are assigned to an afternoon or 11:59 p.m. homework deadline. Results indicate students with an afternoon deadline experience more homework-related stress but have no significant difference in course outcomes. An interpretation is students prefer working late at night and requiring them to turn in assignments earlier creates stress, but this perception does not create a change in study behaviors or results.

2. Literature Review

Homework is a hallmark of the higher education experience, but whether practice outside the classroom helps learning is an often-debated subject (Fernández-Alonso et al. 2017; Cooper 2015). Research has consistently found learning benefits from homework: One experiment showed that between college students randomly assigned to homework or not-homework groups, students who completed homework had higher test scores and grades (Grodner and Rupp 2013). Homework helps students be more active participants in their learning process and can lead to higher educational achievement (Cooper and Valentine 2001; Keith, Diamond-Hallam, and Fine 2004; Kitsantas and Zimmerman 2009). Homework has been shown to cause small but positive increases in academic achievement among math and science students, particularly in the United States (Fan et al. 2017). Weekly assessments help students engage more in virtual learning environments, and more engagement in online work is correlated with better grades (Mogus, Djurdjevic, and Suvak 2012; Holmes 2018).

The amount of homework done electronically has increased in recent years with research results indicating some positive

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impact on learning (Jonsdottir, Bjornsdottir, and Stefansson 2017; Magalhães et al. 2020). Online platforms also offer benefits of more efficient grading for instructors and faster feedback and automated help mechanisms for students (Chow 2015; Palocsay and Stevens 2008). However, online homework platforms have changed the dynamic for when assignments are due, allowing more flexibility for instructors than with pen-and-paper homework that was often due when students came to class. This has opened a new door for research regarding optimal homework deadline times.

Research has documented negative learning effects caused by stress, of which homework is a cause. Homework created stress for 83.3% of students in one survey, more than any other academic, financial, or personal factor (Pitt et al. 2018). Another study of 374 undergraduates found the top three student concerns to be academics, pressure to succeed, and future career plans, and two of those three (academics and pressure to succeed) are potentially affected by homework (Beiter et al. 2015). Promoting effective time management, which is a factor that can be shaped by homework deadline times, is one strategy for improving students' stress and anxiety (Hsu and Goldsmith 2021).

Students who do more hours of homework per night perform better in school but report more mental and physical health problems (Galloway, Conner, and Pope 2013). However, stress before homework deadlines can increase productivity (Bienertova-Vasku, Lenart, and Scheringer 2020), and eustress—positive stress—can predict higher levels of beneficial outcomes like GPA and life satisfaction (Moseley 2018). In reviewing ways to reduce student distress in business courses, King, and Kabat-Farr (2022) note college students face many stressors that build on one another, and they recommend educators design courses with three goals in mind to promote student mental health: simplicity, predictability, and consistency. Part of King and Kabat-Farr's plan for predictability is using consistent deadlines at the same day and time each week. The goal for the instructor, therefore, becomes crafting online homework assignments that foster eustress while mitigating distress, and one tool toward achieving that goal is setting homework deadlines optimally (Chiu, Moss, and Richards 2024).

However, little research has been done about the effect of homework deadline times. Our review of the literature could not identify any studies that experimentally altered homework deadline times to gauge their effect on student outcomes. This study seeks to fill the gap in the research and begin a discussion about deadline times, particularly as it relates to data science and statistics courses.

The issue matters because psychological wellbeing contributes to college students graduating and getting the most out of their college experience. Furthermore, one potential contributor to reaching these goals is setting flexible homework deadlines, and making assignments due during the daytime or early evening may promote sleep (Eblen-Zayas, Burson, and McDermott 2022).

Multiple studies have looked at when online homework is *completed* (rather than when it is due) with results showing students prefer working at night. Ko (2022) found usage peaks between 9 p.m. and 11 p.m., and another analysis of students' online homework activity showed most work is done after lunch with a break around dinner time and continuing until 3 a.m.—except on the night of a deadline, when work stops at midnight (Kortemeyer 2016). Just 28% of online course activities are done during normal business hours, giving universities reason to offer flexible office hours for online faculty so they can answer questions on nights and weekends when students are completing problems (Wu and Hung 2018). An experiment that made some students' assignments due throughout the period between tests and some due the night before an exam revealed no significant performance differences (Raines 2012).

Students indicate homework deadlines are helpful in promoting more consistent study rather than cramming (Richards-Babb et al. 2015), but evidence is mixed on whether homework deadlines should be firm or flexible (meaning students have some control in determining when it is due). Late-night deadlines may contribute to distress because they impose a strict ending time rather than allowing students freedom of unstructured learning time (Rupar and Strong 2021). Miller, Asarta, and Schmidt (2019) experimented with deadline times in an undergraduate economics class on an adaptive online learning system graded based on completion, not accuracy. They assigned students in one section to a weekly 11:30 p.m. Tuesday deadline, and another section of the same course to 11:30 p.m. but only due on the day before each exam. Students with more flexible deadlines performed better, which fits with other research concluding rigid deadlines should be reconsidered for those which give students more choice (Burger, Charness, and Lynham 2011; Bisin and Hyndman 2020).

This research implies students might perform better if given choices to continue working at their natural pace rather than creating an abrupt end at midnight. However, the work must be due sometime, and in one experiment students who set their own assignment deadlines made lower grades than those who had deadlines set for them, indicating a positive effect of imposed deadlines in controlling procrastination (Ariely and Wertenbroch 2002). Likewise, Ford, Broadway, and Mason (2023) found similar results in an experiment involving homework in first-year chemistry classes. Students were given an option to set their own deadlines, take those assigned by the teacher, or not have their homework due until the last day of the semester. The best performance came from those who used the teacher's preset deadlines, suggesting students often lack motivation to complete homework on their own. In that sense, a deadline imposes boundaries that motivate more studying.

These findings point to how the psychology of procrastination plays an important role in the strategy of setting homework times. Students who procrastinate report online learning is less effective overall, and procrastination is negatively associated with the characteristics that make homework effective like task strategy, emotion control, and environmental structure (Hong, Lee, and Ye 2021). Procrastinating students tend to have lower course grades, giving instructors incentive to structure homework to minimize the possibility of students waiting until the last minute (Nieberding and Heckler 2023). The sooner students complete homework after a lecture, the higher their grade on exams, and this finding is particularly strong for struggling students at risk of failing a course (Cosio and Williamson 2019). A study of 20 courses showed a high, positive correlation between time before the deadline when the student submits an assign-

ment and course grades (Wang et al. 2023). Students who access online learning resources consistently do better than those who only engage with it shortly before the deadline (Knight 2010). This evidence may lend support to setting homework deadlines during the afternoon; it is hypothesized that afternoon deadlines would encourage incremental progress because the assignment could not be done all at once on the due date because of obligations throughout the day. The work would need to be done earlier and with more engagement throughout the week, rather than waiting until the night it is due.

Why then do students indicate preference for a midnight deadline if it potentially increases distress and lowers academic performance? Rupar and Strong (2021) postulate it's primarily because a midnight deadline gives consistency between all classes for when assignments are due, reducing complexity rather than contributing to long-term learning. However, students may simply prefer to work at night (Ko 2022; Kortemeyer 2016).

Instructors often structure homework to encourage students to complete it sooner and avoid the negative consequences of procrastination, but this creates a tension with student expectations if students are accustomed to 11:59 p.m. deadlines. Disrupting the consistency of the deadlines might carry more negative effects by increasing student stress because they are surprised by the unusual afternoon deadlines than the positive effects of avoiding procrastination. This study seeks to contribute to the literature by exploring how changing homework deadlines affect learning outcomes and student stress levels.

3. Methodology

The research was conducted at a private, four-year university in the United States in a 200-level applied statistics course, Quantitative Business Analysis, which is required for all business majors. Course topics include descriptive statistics, probability distributions, sampling, statistical inference, regression analysis, and spreadsheet modeling, all with a focus in business applications. The same instructor taught four sections during a calendar year: two in the spring semester and then two in the fall semester. The syllabi and class lectures remained the same in all the sections, which were held on Mondays, Wednesdays, and Fridays. Courses were offered in the spring at 8:30 a.m. and 9:30 a.m., and 8:30 a.m. and 12:30 p.m. in the fall. Students are generally traditional, on-campus undergraduates majoring in business.

In the first semester of the study, section 1 had homework deadlines set at 4 p.m., while the deadline for section 2 was 11:59 p.m. The university's student mental health counselors suggested 4 p.m., saying it would allow students to complete homework before nonacademic campus activities began, thus, freeing them to enjoy social time without worrying about deadlines.

Nine assignments, each covering a chapter in the textbook, were posted at the beginning of the 16-week semester along with their deadlines, and students could complete the homework at any time. Problems required statistical analysis on spreadsheets to answer questions on an automatically graded online platform. The deadlines came a week after the material was covered in class. Points were automatically deducted from late homework assignments at a rate of 20% per day, an incentive intended to encourage timely submissions. Students could request extensions on homework before the deadline passed, and such extensions were granted at the instructor's discretion. However, the syllabus noted that extensions would not be granted for routine reasons such as poor planning or regular conflicts like work and family vacations. Most students completed the assignments during the week between when they were assigned and the due date, although one student worked ahead of the course lectures.

Concerns existed that the time the course started would confound the results because students in the two sections had differing amounts of time between when their class period ended and when their homework was due. To mitigate this issue, the study was replicated during the fall semester with the deadline time flipped: The earlier section was assigned an 11:59 p.m. deadline and the later section 4 p.m.

The goal was to limit the difference between the sections to only homework due times. Efforts were made to keep other factors about the course constant, including the textbook, lectures, and homework platform and questions. In the spring, section 1 had an enrollment of 31 and section 2 had 30 students. In the fall, section 1's enrollment was 24 and section 2's was 23. One student failed the course the first semester and retook it the second semester; this student's survey results are included in both semesters, and they were assigned to the midnight deadline group both times.

Table 1 shows demographic characteristics of students who completed course surveys and the differences between the afternoon and midnight deadline cohorts. Half of the students were in their second-year, with another 30% in their third year. A Chi-square test for homogeneity was conducted to compare the distributions of student-year levels across the two homework deadline groups. Results indicated no statistically discernible difference in the distribution of year levels, $\chi^2(3, N = 84) = 5.98, p = 0.11$.

The mean GPAs (3.46 and 3.37) and ACT scores (25.7 and 24.6) between students with the different deadline times were not statistically different based on t-tests with variances assumed unequal ($p = 0.44$ and $p = 0.23$, respectively). Likewise, there was no statistically discernible difference between grades on the homework and for the course between the two cohorts ($p = 0.94$ and $p = 0.63$, respectively). Figure 1 shows a comparison by the two deadline times of the grade distributions, which are similar.

Although the percentage of students identifying as female in the courses was 23% for the afternoon group and 39% for the midnight group, the difference was not statistically significant based on a two-sample test for equality of proportions, $\chi^2(1, N = 84) = 1.76, p = 0.19$). Overall, an analysis of the demographic characteristics of the two groups supports that the quasi-experiment was successful in creating two groups that were otherwise similar apart from the different homework deadlines.

The university's Institutional Review Board approved surveying students, and participation in the study had no effect on student grades or assignment times. Those who opted in completed a seven-question survey administered on paper at the end of the semester and did not receive any incentive to do so

Table 1. Student demographic characteristics, overall and by homework deadline time.

	All students Mean (SD), n	Afternoon deadline Mean (SD), n	Midnight deadline Mean (SD), n
Homework minutes	960.9 (519.3), n = 84	985.6 (592.4), n = 41	935 (435.6), n = 41
Female	30.9%, n = 26	23.3%, n = 10	39%, n = 16
GPA	3.42 (0.54), n = 84	3.37 (0.49), n = 43	3.46 (0.59), n = 41
ACT	25.0 (4.3), n = 67	24.3 (3.8), n = 33	25.7 (4.8), n = 34
Year in school	n = 84	n = 43	n = 41
First-year	8.3%, n = 7	2.3%, n = 1	14.6%, n = 6
Second-year	50%, n = 42	46.5%, n = 20	53.7%, n = 22
Third-year	29.8%, n = 25	37.2%, n = 16	21.9%, n = 9
Fourth-year	11.9%, n = 10	14%, n = 6	9.8%, n = 4
Grade homework	82.2% (17%), n = 84	82.3% (16%), n = 43	82% (18%), n = 41
Grade course	88.8% (10%), n = 84	89.3% (9%), n = 43	88.2% (10%), n = 41

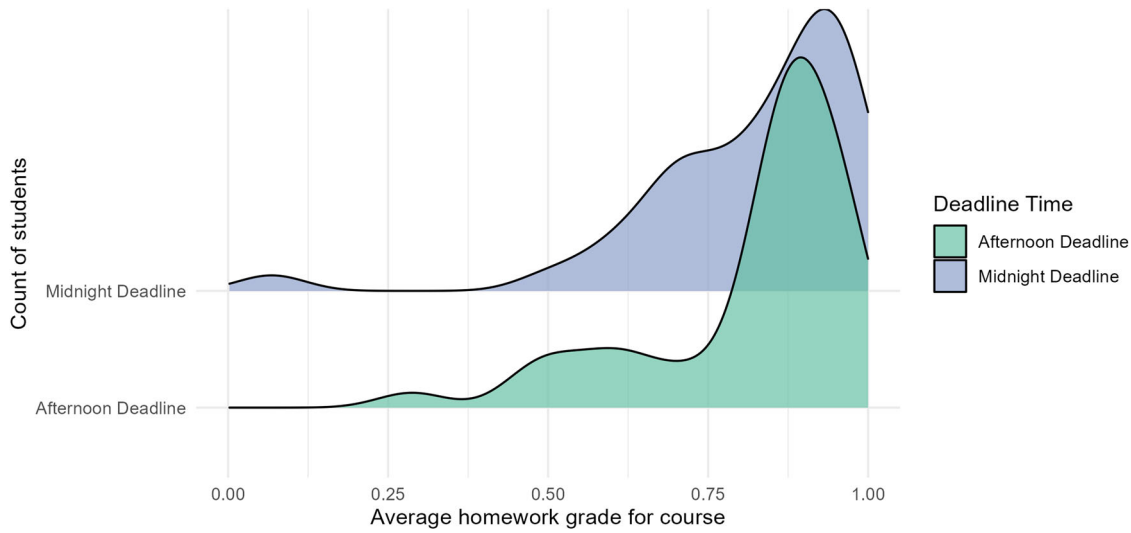

Figure 1. Distribution of homework grades by deadline time.

like bonus points. Overall, 41 usable surveys were completed by students with an 11:59 p.m. homework deadline, and 43 usable surveys with a 4 p.m. deadline.

Three of the questions used a five-point scale (see Figure 2):

- How effective do you believe the homework was in helping you learn the course material?
- How much did your homework deadline time and date affect your ability to get it done on time?
- How much did the homework in this class affect your stress level throughout the semester?

Three other questions asked for numerical answers:

- How much time do you estimate you spent on average per homework assignment?
- How many homework extensions did you request in the course?
- How many times did you turn in a homework assignment late and not request an extension?

A final question asked what time students preferred homework to be due with six categories: weekday afternoon, weekday midnight, weekend afternoon, weekend midnight, just before the class period starts, and "other," which allowed students to write in another time.

The question that said, "How much did your homework deadline time and date affect your ability to get it done on time?" included five options on a Likert scale labeled as very harmful, somewhat harmful, neutral, somewhat helpful, and very helpful. The intent for this question was to gauge how the earlier or later deadline times impacted timeliness of student submissions. An earlier deadline time might result in making it more difficult for students to get assignments done on time simply because they had fewer hours to complete it, but the earlier time might also encourage avoiding procrastination and result in getting assignments done earlier.

An analysis of covariance (ANCOVA) model was used to evaluate the effects on the outcome variables associated with the categorical predictor variables while controlling for other covariates and interactions between the predictor variables (Hair et al. 2019). Three ANCOVA models were run for the three outcomes of interest—homework effectiveness, homework timeliness, and stress—as measured by the first three questions on the survey. Each model had the same independent variables and covariates. Independent variables were the deadline time, gender, and year in school, and the covariates were student GPA and minutes spent doing homework, as measured by the online homework software. This setup helps isolate differences between the two groups associated with the different deadlines while controlling for student academic

Survey – Perceptions of homework times – Quantitative Business Analysis, spring 2022

Name:

These questions apply only to homework done through the Pearson website; do not consider in-class problems when answering.

1. How effective do you believe the homework was in helping you learn the course material?
    1. Very ineffective
    2. Somewhat ineffective
    3. Neutral
    4. Somewhat effective
    5. Very effective
2. How much did your homework deadline time and date affect your ability to get it done on time?
    1. Very harmful
    2. Somewhat harmful
    3. Neutral
    4. Somewhat helpful
    5. Very helpful
3. How much did the homework in this class affect your stress level throughout the semester?
    1. Made it much worse
    2. Made it somewhat worse
    3. Neutral
    4. Made it somewhat better
    5. Made it much better
4. How much time do you estimate you spent on average per homework assignment?

    _____
5. What time do you prefer that homework be due?
    1. Weekday afternoon
    2. Weekday midnight
    3. Weekend afternoon
    4. Weekend midnight
    5. Just before the class period starts
    6. Other – If prefer other, please state when: _____
6. How many homework extensions did you request in the course?

    _____
7. How many times did you turn in a homework assignment late and not request an extension?

    _____

Figure 2. Survey – Perceptions of homework times – Quantitative Business Analysis, spring 2022.

achievement (as measured by GPA) and effort (as measured by hours spent on the homework). This approach is common on quasi-experiments in education involving different classrooms (Conway et al. 2019).

In addition to the surveys, course evaluations were also used to gauge student sentiment about the homework. The evaluations are administered by the university's Office of Institutional Research at the end of each semester, and students completed them anonymously via a web form. To quantify the text data contained in the written comments, sentiment analysis was performed using the Valence Aware Dictionary and sentiment Reasoner (VADER) within the R statistical software package. VADER is designed for sentiments expressed in social media settings, and it is useful for other applications as well, including course evaluations where students tend to use informal language (Hutto and Gilbert 2014). It uses a dictionary-based method with sentiment scores assigned to words on a -4 – 4 scale based on how negative or positive the words are. Sentiment scores are then calculated based on word sentiment values and a series of grammatical rules that consider context (Hutto and Gilbert 2014). With this limited number of texts that are fairly short, human sentiment scoring could have been used. However, sentiment analysis performed with tools like VADER are more standardized and could more easily be reproduced and compared in future studies that might analyze longer sections of text. Also, Hutto and Gilbert (2014) demonstrated VADER outperforms individual human raters in common sentiment analysis tasks.

4. Results

The results point to two key findings: First, students in the study indicated they prefer later homework deadlines and that later

Table 2. Student survey results, overall and by homework deadline time.

	All students Mean (SD), n	Afternoon deadline Mean (SD), n	Midnight deadline Mean (SD), n
Homework effectiveness	4.05 (0.96), n = 84	3.93 (1.12), n = 43	4.17 (0.74), n = 41
Deadline effectiveness	3.43 (1.11), n = 84	2.98 (1.14), n = 43	3.9 (0.86), n = 41
Homework stress	2.81 (0.83), n = 84	2.65 (0.840), n = 43	2.98 (0.79), n = 41
Extensions	0.49 (0.91), n = 83	0.56 (0.98), n = 43	0.42 (0.84), n = 40
Late submissions	0.71 (1.02), n = 82	0.8 (1.1), n = 43	0.6 (0.93), n = 39
Preferred deadline	n = 84	n = 43	n = 41
Weekday afternoon	2.4%, n = 2	4.7%, n = 2	0%, n = 0
Weekday midnight	34.5%, n = 28	30.2%, n = 13	36.6%, n = 15
Weekend afternoon	2.4%, n = 2	0%, n = 0	4.9%, n = 2
Weekend midnight	46.4%, n = 38	48.8%, n = 21	41.5%, n = 17
Just before class	11.9%, n = 10	11.6%, n = 5	12.2%, n = 5
Other	2.4%, n = 4	4.7%, n = 2	4.9%, n = 2

Table 3. ANCOVA summary results for the timeliness model.

	Degrees of freedom	Sum of Squares	Mean Square	F	P	Partial η squared
Midnight deadline	1	17.98	17.98	17.59	0.00	0.158
Female	1	1.02	1.02	0.99	0.32	0.017
Year in school	3	8.01	2.67	2.61	0.06	0.082
GPA	1	0.93	0.93	0.91	0.34	0.004
Homework minutes	1	1.24	1.24	1.22	0.27	0.019
Midnight deadline and female	1	1.73	1.73	1.70	0.20	0.020
Midnight deadline and year in school	3	1.13	0.38	0.37	0.78	0.016
Female and year in school	3	1.06	0.35	0.35	0.79	0.015
Midnight deadline and female and year in school	2	0.98	0.49	0.48	0.62	0.014
Residuals	67	68.49	1.02			

homework deadlines were associated with less stress. Second, the stress associated with earlier homework deadlines did not negatively affect academic outcomes, as measured by student grades in the course and on the homework assignments.

Table 2 summarizes the results from the student surveys, broken down by the assigned deadline time. Homework effectiveness is from the question asking, "How effective do you believe the homework was in helping you learn the course material?" Deadline effectiveness is from the question, "How much did your homework deadline time and date affect your ability to get it done on time?" Stress is from the question, "How much did the homework in this class affect your stress level throughout the semester?"

The largest mean difference between the groups was for deadline effectiveness (based on the survey question, "How much did your homework deadline time and date affect your ability to get it done on time?"). The midnight deadline group gave a mean score of 3.9 compared to 2.98 for the afternoon group, $t(78) = 4.2, p < 0.001$. Students reported the later deadline helped them complete assignments on time better than the afternoon time.

To further explore this relationship, an ANCOVA model with deadline effectiveness as the dependent variable was implemented. The results again showed a statistically discernible difference for the 11:59 p.m. and the afternoon deadline groups when controlling for the covariates and interactions (see Table 3). The partial η squared was 0.158, indicating a large effect size (Cohen 1969; Richardson 2011).

Also, the question about which deadline time students preferred showed a sizeable majority (81%) selected midnight (either on a weekday or weekend) as their preferred homework due time. Only 5% preferred an afternoon deadline, while 12% wanted deadlines just before the class period begins. There was slight preference in favor of weekend deadline times, with 40 respondents (48%) wanting a weekend time compared to 30 (36%) wanting a weekday time. Clearly, the deadline time mattered to the students, and they strongly preferred midnight.

Regarding stress, the ANCOVA model's dependent variable was based on the survey question asking how much the homework affected their stress level throughout the semester, with 1 being "made it much worse" and 5 being "made it much better." Based on deadline time, the difference between the group means was significant at the 90% level, with students with the afternoon deadline reporting higher stress because of the homework (see Table 4). Although not a large effect, which is perhaps not surprising considering college students face many other significant stressors, the results point toward a negative effect on stress by the earlier deadline.

The largest partial η squared value for the ANCOVA stress model was for the interaction between deadline and year in school. About 8.5% of variation in stress was associated with that variable. Looking more closely at the descriptive statistics broken down by year in school reveals that although most students reported the midnight deadline caused more stress, fourth-year students reported less stress from the afternoon deadline. The mean response to the stress question was 3.2 for fourth-year students with afternoon deadlines compared to 2.5 for fourth-year students with midnight deadlines. A possible interpretation is more experienced students are better equipped to manage their time and are better able to adjust to the earlier deadline. Although the sample size for fourth-year students (n= 10) is too small to draw firm conclusions from, the relationship between student experience and homework deadline stress could be explored further in future studies.

Table 4. ANCOVA summary results for the stress model.

	Degrees of freedom	Sum of Squares	Mean Square	F	P	Partial η squared
Midnight deadline	1	2.21	2.21	3.53	0.06	0.043
Female	1	0.26	0.26	0.41	0.52	0.011
Year in school	3	1.12	0.37	0.60	0.62	0.007
GPA	1	2.34	2.34	3.75	0.06	0.032
Homework minutes	1	1.98	1.98	3.17	0.08	0.034
Midnight deadline and female	1	0.48	0.48	0.77	0.38	0.011
Midnight deadline and year in school	3	3.61	1.20	1.92	0.13	0.085
Female and year in school	3	1.46	0.49	0.78	0.51	0.033
Midnight deadline and female and year in school	2	1.57	0.78	1.25	0.29	0.036
Residuals	67	41.93	0.63			

Table 5. ANCOVA summary results for the homework effectiveness model.

	Degrees of freedom	Sum of Squares	Mean Square	F	P	Partial η squared
Midnight deadline	1	1.21	1.21	1.30	0.26	0.014
Female	1	0.06	0.06	0.07	0.80	0.007
Year in school	3	5.04	1.68	1.80	0.16	0.075
GPA	1	1.11	1.11	1.19	0.28	0.006
Homework minutes	1	0.32	0.32	0.35	0.56	0.004
Midnight deadline and female	1	0.03	0.03	0.04	0.85	0.001
Midnight deadline and year in school	3	2.74	0.91	0.98	0.41	0.039
Female and year in school	3	0.94	0.31	0.33	0.80	0.015
Midnight deadline and female and year in school	2	1.69	0.85	0.90	0.41	0.026
Residuals	67	62.65	0.94			

Table 6. ANCOVA summary results for the homework grade model.

	Degrees of freedom	Sum of Squares	Mean Square	F	P	Partial η squared
Midnight deadline	1	0.00	0.00	0.00	0.94	0.000
Female	1	0.04	0.04	1.46	0.23	0.013
Year in school	3	0.08	0.03	0.88	0.46	0.030
GPA	1	0.05	0.05	1.52	0.22	0.021
Homework minutes	1	0.00	0.00	0.05	0.82	0.001
Midnight deadline and female	1	0.00	0.00	0.00	0.98	0.004
Midnight deadline and year in school	3	0.02	0.01	0.25	0.86	0.022
Female and year in school	3	0.05	0.02	0.60	0.62	0.026
Midnight deadline and female and year in school	2	0.04	0.02	0.58	0.56	0.017
Residuals	67	2.06	0.03			

But while students may have *perceived* the 11:59 p.m. deadline made it easier to complete assignments on time, did it make a difference in their actual learning? The results suggest no in this case: Students did not spend a statistically discernible different amount of time on the homework, make different grades, turn in more assignments late, or ask for more extensions. Also, the ANCOVA models using dependent variables of the homework effectiveness survey question and the student homework grades did not show any statistically discernible differences between groups based on deadline time, gender, year in school, and interactions between those variables (see Tables 5 and 6).

These results indicate students put the same amount of effort into the class and experienced similar outcomes, regardless of which homework deadline they were assigned. Cognitive dissonance appears to be at work: How students feel about the deadline times differs from their actions regarding the homework times. On one hand, this may be evidence that, at least in this situation, teachers could not affect course outcomes by changing homework deadlines.

But looking at the results another way, it gives freedom for further experimentation regarding homework deadline times: Even if students do necessarily like homework to be due at times other than 11:59 p.m., trying different times is unlikely to negatively hurt their learning. In that case, other attempts could be made, in light of the problem psychologists have identified with students staying up too late, to find other homework deadline schemes that may improve performance.

To get another perspective on student perception of the deadline times, comments from the end-of-course evaluations were analyzed from both a qualitative and a quantitative perspective. In a qualitative analysis, all the references to homework deadlines in the comments were reviewed. Seven students who had the afternoon deadline mentioned the homework deadlines, and all of their remarks were negative. For example, one student said future courses should change the time to midnight because "that odd due date is easy to miss." In contrast, students with a midnight deadline did not make any comments regarding the homework deadlines in the course evaluations. This suggests the standard deadline did not create any noteworthy problems.

Sentiment analysis was then conducted to quantify students' opinions contained in the evaluations. The percentage of positive, negative, and neutral words were calculated for students

from both deadline groups, as well as an overall compound valence score. A compound score is made on a $-1 - 1$ scale where more negative values reflect more negative sentiment and more positive values reflect more positive sentiment.

For all the comments made by the midnight deadline group, the mean compound score was .261, compared to .246 for the afternoon deadline. However, the discernible difference in means was not statistically significant, $t(182) = -0.28$, $p = 0.78$. A college course includes many other aspects that students are likely to comment on in addition to homework deadlines, and, therefore, it is not surprising that overall sentiment about the course is similar between the two groups. However, the afternoon group was more than twice as likely to express negative sentiment: The negative score, reflecting the weighted percentage of negative words in the text, was 6.73% for the afternoon group versus 3.01% for the midnight group, $t(157) = 2.1$, $p = 0.037$. When analyzed on a question-by-question basis for each of the four text-based questions on the course evaluations, the same trend applied. For all four questions, the percentage of negative words was higher for the afternoon deadline. This sentiment analysis aligns with the survey results showing students did not like the earlier deadlines.

One concern about the validity of the results is students who completed the homework well ahead of the deadlines are not truly in the population of interest and may be distorting the results. To assess their impact on the results, the final login time into the online homework system was checked for each student in the classes. Only three students last logged into the software more than a week before the last assignment was due. Two of those students withdrew from course early, well before the surveys were administered, and the third student, who completed all the homework assignments within the first two weeks of the course, opted not to participate in the study. Even though students had the option of working ahead on homework, only one student appears to have actually done so, and their results are not included because they opted out of the research.

5. Discussion and Conclusion

Little previous research has been done on the relationship between homework deadline times and academic outcomes, and this study attempts to open a door of inquiry into this area that holds potential to impact student learning and wellbeing. The findings point toward students in this study disliking earlier deadlines, which is in line with existing research. They perceive earlier deadlines as harming their ability to get work done on time and increasing their stress levels as a result.

However, this extra time did not translate into different course outcomes; students in the midnight group spent similar amounts of time on homework and made nearly identical grades on average, compared with the afternoon group.

These results provide two potentially contradictory insights for instructors. On one hand, teachers may want to be cautious in changing homework deadlines from the accepted standard of 11:59 p.m. because it could lead to student dissatisfaction and more negative course evaluations. Because the choice of homework time has little impact on instructors, who can grade assignments whenever they wish and opt to use automated grading software, it would seemingly make little sense to make a change that is unlikely to result in better outcomes while at the same time provoking student discontent. In this sense, allowing students to complete the homework at the time fitting their preferences—later at night—would appear to be the better choice over creating earlier deadlines to encourage more sleep. Perhaps students are rational economic agents who know best for themselves when they are most effective at completing homework, and the best course of action in that case would be to allow them freedom to choose.

However, because the actual course outcomes did not seem to change, it might not be detrimental to students to experiment with different homework deadline times. This study was conducted with just two different homework times (4 p.m. and 11:59 p.m.) for one business course at one university. It would be helpful to consider alternative times, which might have a different effect on course outcomes. Also, the study was conducted among traditional on-campus undergraduates and may not generalize to other universities that feature more commuters, first-generation students, or adult learners.

Future research could change homework deadline times at the department, college, or university levels to test the effects of changing homework deadlines for multiple or all courses. For example, a potential mental health benefit from earlier homework deadlines is encouraging college students to go to bed earlier. Homework in just one class may not be enough to alter students' sleep habits but changing it for all of a students' courses for a semester might have a more substantial effect. Also, would different variations of homework deadlines, perhaps farther apart or at more starkly different times, create different outcomes? At the very least, the results of this study indicate such experimentation comes with a low risk of negatively affecting student learning, and it could lead to insights that could help alleviate student stress, which is an important issue for higher education.

In a similar vein, future research could explore how homework deadlines affect sleep. This study did not ask students about the effects of homework on sleep, but the relationship between those two variables would be worthy of exploring, particularly since the de facto 11:59 p.m. time may discourage healthy rest routines. Another area worth exploration would be the relationship between undergraduates' year in school and homework deadlines. The data here, although limited by a small number of fourth-year students ($n = 10$), suggests more experienced students are better able to adapt to afternoon deadlines and might benefit from them more than students still getting used to college.

The amount of notice students have before homework is due also might matter. For example, students in this quasi-experiment had all homework assignments available from the first day of the semester. Most students waited until the week the homework was due to complete it, which was after the material was presented in class, but they had the option of doing it earlier. In fact, one student completed all the homework assignments in the first two weeks of the course because it was within a two-week free trial period for the homework software offered by the textbook provider. Regardless, if students had less time to complete homework, changing the deadlines might have a greater effect on learning outcomes.

This study creates a starting point for a discussion about homework deadline times, which increasingly matters as more college work assignments are completed online and as a student mental health crisis continues with homework being college students No. 1 stressor (Pitt et al. 2018).

Supplementary Materials

The supplementary files include the R file and all the data files which are used in it.

Disclosure Statement

The author reports there are no competing interests to declare.

Data Availability Statement

The deidentified data and code that support the findings of this study are openly available at the Open Science Framework (OSF): *https://osf.io/u2efq/*.

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